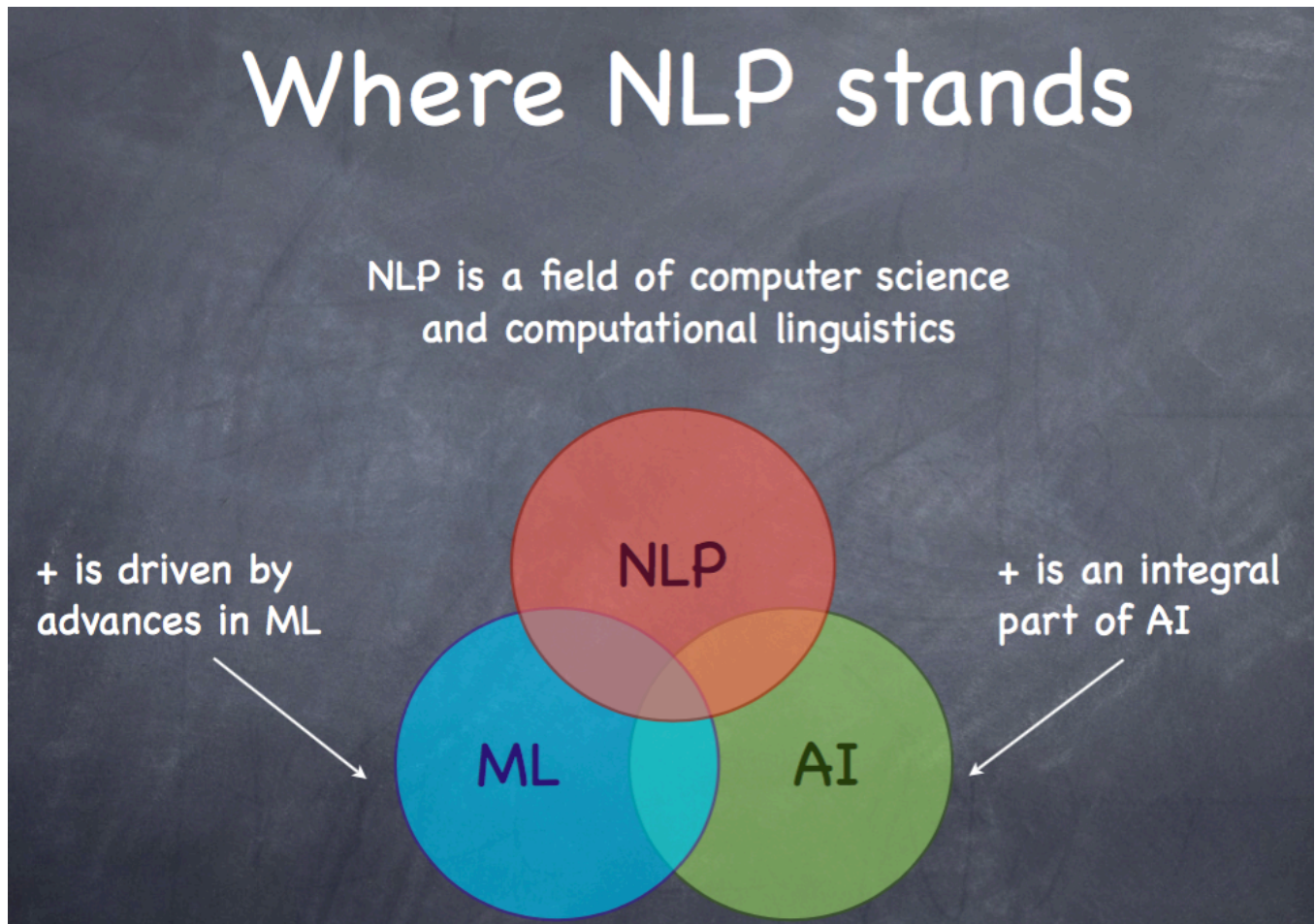


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Introduction to Natural Language processing

Natural language processing (NLP): Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.



What is it all about?

What's that all about

Let's start with the basics:



we speak



we read



we write

all that using language

But that's not all...



- we think of the world around us
- we dream
- we make decisions and plans
- all in natural language, i.e. in words

What are the raw materials for NLP revolution?

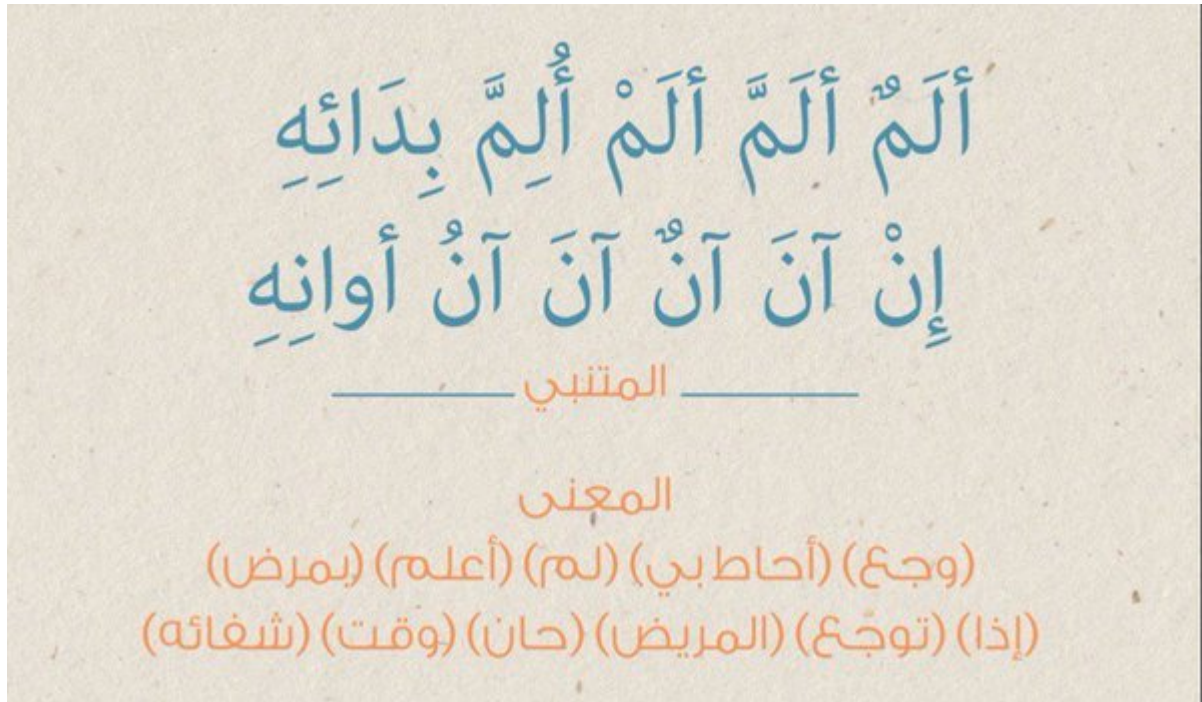
1. **Data:** There is a huge amount of human knowledge that has been digitized.
2. **Compute:** Our personal computers are very powerful, and it has never been easier to rent more compute power.
3. **Algorithms:** We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

Why NLP is hard?

It's the nature of the human language that makes NLP difficult.

Some examples:

1. **Ambiguity:** One word/sentence can have different meanings.



2. Spelling Errors: There are some words that are usually misspelled by people.



Top 10 Misspelled Words

separate ✓	seperate ✗
calendar ✓	calender ✗
unnecessary ✓	unecessary ✗
questionnaire ✓	questionaire ✗
accidentally ✓	accidentaly ✗
necessary ✓	neccessary ✗
surprise ✓	suprise ✗
embarrass ✓	embarass ✗
accommodation ✓	accomodation ✗
receive ✓	recieve ✗

3. Phonetics and Phonology: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

4. Order of Context: "Dog bites man" and "Man bites dog" have vastly different meanings

In ML terms, why is NLP hard?

- 1. Input:** Large, discrete, open-ended input state spaces
- 2. Output:** Variable length output spaces
- 3. Algorithms:** Recycled from other fields

Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF
- DOCX

NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

NLTK

Natural Language Toolkit (nltk) is a Python package for NLP <https://www.nltk.org/> (<https://www.nltk.org/>)

What are Corpus / Corpora?

a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
%pip install nltk pandas
```

```
Requirement already satisfied: nltk in /Users/almadani/Desktop/nlp
lab/.venv/lib/python3.13/site-packages (3.10.3)
Requirement already satisfied: pandas in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (3.0.0)
Requirement already satisfied: defusedxml in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
nltk) (0.7.1)
Requirement already satisfied: click in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
nltk) (8.4.2)
Requirement already satisfied: joblib in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in
/Users/almadani/Library/Python/3.13/lib/python/site-packages (from nltk) (2026.7.19)
Requirement already satisfied: tqdm in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
nltk) (4.67.3)
Requirement already satisfied: numpy>=1.26.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
pandas) (2.4.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/almadani/Library/Python/3.13/lib/python/site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: six>=1.5 in
/Users/almadani/Library/Python/3.13/lib/python/site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
```

[notice] A new release of pip is available: 25.2 -> 26.2.1

[notice] To update, run: /Users/almadani/Desktop/nlp lab/.venv/bin/python -m pip
install --upgrade pip

Note: you may need to restart the kernel to use updated packages.

```
import nltk
```

Arabic Libraries (Extra)

Camel-Tools: <https://camel-tools.readthedocs.io/en/latest/> (<https://camel-tools.readthedocs.io/en/latest/>)

PyArabic: <https://pyarabic.readthedocs.io/ar/latest/> (<https://pyarabic.readthedocs.io/ar/latest/>)

Farasa: <https://farasa.qcri.org/> (<https://farasa.qcri.org/>)

NLP Applications

1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

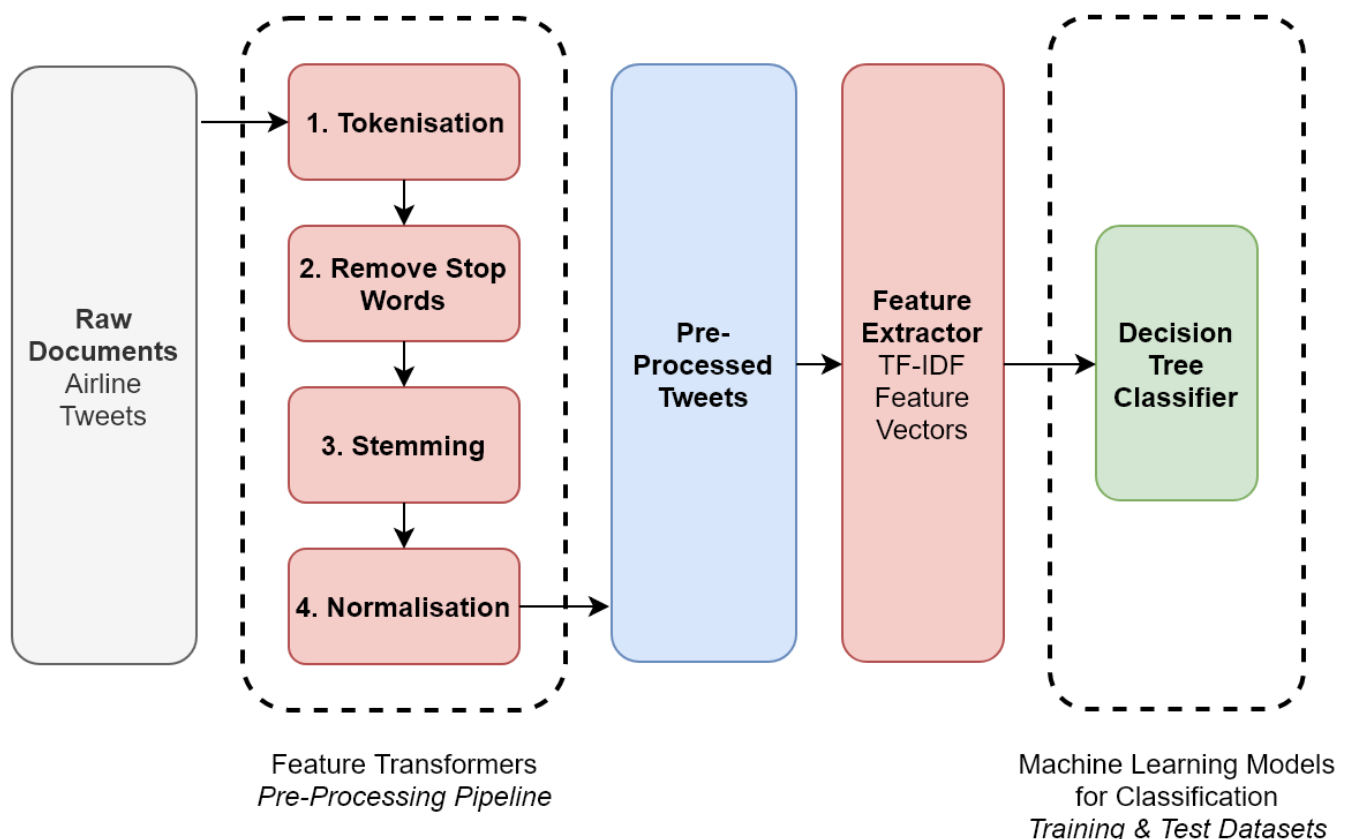
2. Machine Translation

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

3. Sentiment Analysis

Identifying sentiments and opinions stated in a text.

Example of an NLP Pipeline



1. Text Acquisition:

Obtain the text data from a source, such as a file, a web page, or a database.

2. Text Preprocessing:

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all tokens to lowercase to ensure uniformity.
- **Removing Punctuation:** Strip away punctuation marks from tokens.
- **Stopword Removal:** Eliminate common words (e.g., "and," "the," "is") that may not carry significant meaning.
- **Stemming:** Removes prefixes/suffixes to get a root form. (Running -> run)
- **Lemmatization:** Converts word to base/dictionary form (lemma). (better -> good).

3. Feature Extraction:

- **Bag-of-Words (BoW):** Create a vector representation of the text using the frequency of each token in the document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Assign weights to tokens based on their importance in the document relative to the entire corpus.

4. Model Building:

Choose a machine learning model for sentiment analysis, such as a Naive Bayes classifier, Support Vector Machine (SVM), or a neural network.

Task#1:

Students should work in groups and search online for one of the different types of datasets. • CSV • Text Files • JSON • SQL • HTML • PDF • DOCX

Task#2:

Download and open the dataset.

```
%pip install kagglehub
```

```
Requirement already satisfied: kagglehub in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (1.0.1)
Requirement already satisfied: kagglesdk<1.0,>=0.1.22 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
kagglehub) (0.1.23)
Requirement already satisfied: packaging in
/Users/almadani/Library/Python/3.13/lib/python/site-packages (from kagglehub) (26.0)
Requirement already satisfied: pyyaml in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
kagglehub) (6.0.3)
Requirement already satisfied: requests in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
kagglehub) (2.34.2)
Requirement already satisfied: tqdm in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
kagglehub) (4.67.3)
Requirement already satisfied: protobuf in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
kagglesdk<1.0,>=0.1.22->kagglehub) (7.35.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
requests->kagglehub) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
requests->kagglehub) (3.16)
```

Requirement already satisfied: urllib3<3,>=1.26 in
 /Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
 requests->kagglehub) (2.7.0)
 Requirement already satisfied: certifi>=2023.5.7 in
 /Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from
 requests->kagglehub) (2026.5.20)

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 install --upgrade pip

Note: you may need to restart the kernel to use updated packages.

```
import kagglehub
from pathlib import Path
from zipfile import ZipFile

folder = Path('lab1') if Path('lab1').is_dir() else Path('.')
if (folder / 'IMDB Dataset.csv.zip').exists():
    with ZipFile(folder / 'IMDB Dataset.csv.zip') as archive:
        archive.extract('IMDB Dataset.csv', folder)
        path = str(folder)
else:
    path = kagglehub.dataset_download("lakshmi25npathi/imdb-dataset-of-50k-movie-reviews")

print("Path to dataset files:", path)
```

Path to dataset files: .

/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
 packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and
 ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
 from .autonotebook import tqdm as notebook_tqdm

```
import pandas as pd

# load the CSV file in a dataframe
df = pd.read_csv(Path(path) / "IMDB Dataset.csv")

# Show the first few rows
df.head()
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

```
df.describe()
```

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's show!!! It was a variety and not...	positive
freq	5	25000

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    review    50000 non-null   str
1    sentiment  50000 non-null   str
dtypes: str(2)
memory usage: 63.6 MB
```

```
# Check class distribution
print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```